Caner Derici, PhD

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Technical Skills

Areas of Expertise:	Compilers & Programming Languages · Machine Learning · Distributed Systems
Languages:	C++ · LLVM · MLIR · Go · Python · Racket/Scheme · Java · SQL · JavaScript
Workflows & Productivity:	Linux · Neovim · tmux · VSCode · Copilot · Git · GH Actions · Obsidian · Toggl · Todoist
API, DB & Misc:	REST · gRPC · OpenAPI · FastAPI · DQLite · MongoDB · PostgreSQL · CI/CD · Jenkins

Education

Ph.D., Indiana University, Bloomington, Computer Science, Compilers & Programming Languages — 2015 – 2025

Dissertation: Self-Hosting Functional Programming Languages on Meta-Tracing JIT Compilers

M.Sc., Boğaziçi University, Computer Science, Machine Learning — 2012 – 2014

B.Sc., Bilgi University, Computer Science — 2005 – 2010

Selected Projects

Pycket: A tracing JIT compiler for full-scale bootstrapping functional language

Built the first tracing JIT capable of running a self-hosting language on a meta-tracing JIT backend, and worked on it for over five years during PhD. Created a new IR and implemented performance analysis tools, run-time optimizations, and formalisms to improve throughput. Added code-gen for the FFI layer, engines, and meta-continuations for user threads. Used Python, C++, and Racket.

Around the World in 26 Languages

Building a personal collection of experimental compilers, one for each letter of the alphabet, named after a world city. Each explores different ideas in language design, IR construction, optimizations, and run-time. Targeted backends include LLVM IR, MLIR, NVPTX for CUDA. All in C++.

Racket IR: Linklets

Contributed to Linklets, the intermediate representation and compilation unit format used in current Racket releases on Chez (see publications). Documented Linklets' design and developed its formal semantics in my PhD dissertation.

Rax: A full-stack nanopass compiler from Racket to x86_64

Implemented all the passes (closure conversion, register allocation, code-gen, etc.), along with garbage collection. Developed optimizations, such as inlining, loop-invariant code motion, and proper tail calls. In Racket, and C++.

FARS: Functional Automated Reasoning System

A resolution/refutation theorem prover, for expressions in first-order predicate logic with equality. Used binary paramodulation, and forward and backward subsumption for equational deduction. In Racket.

HazirCevap (Witty): A closed-domain question answering system for high school students

Government-funded large-scale question-answering system. M.Sc. thesis on NLP. Led the R&D team (3 faculties, 4 grad students). Developed a Hidden Markov Random Field model for question analysis, and relevance metrics for information retrieval and response generation (see publications). In Python, and JavaScript.

Experience

Canonical USA — Remote, US

SWE II (L4 - IC3) · Domain Specific Language Compiler & Distributed Orchestration — *2021 – 2024*

- Designed and developed a custom scripting DSL for (CPU & memory) restricted computations. Built compiler on top of a modified Lua interpreter in Go.
- Developed and maintained Juju, an eventually consistent distributed orchestration system used by ~200 companies globally, capable of handling 1000+-node workloads with 99.9% availability on any infrastructure (Kubernetes or otherwise) across various cloud providers (e.g., AWS, GCE). All in Go.
- Improved reliability and fault tolerance by implementing edge machine services on relational DQLite back-end, migrating from NoSQL MongoDB (e.g., sample PR).
- Owned client libraries for three years–python-libjuju, Terraform Juju Provider; doubled active users and maintained a steady release cadence.
- Took part in roadmap planning, coordinated cross-team work; mentored junior engineers, and improved hiring by creating a structured bootcamp process that cut down onboarding new engineers from 6 months to 1 month.

Indiana University — IN, US

Research Assistant, Course Instructor — *2015 – 2021*

- Independently took an ambiguous, uncharted compiler problem to a working product; built the first-ever tracing JIT compiler that is a full-scale runtime for a self-hosting, production-grade language. Conducted a full performance investigation and designed new optimization algorithms (see PhD dissertation).
- Taught data structures & algorithms, compilers, virtual machines, and domain specific languages.

Asseco SEE Group

Software Engineer *2010-2012*

- International software company developing virtual payment systems for e-commerce platforms. I developed and delivered 3 virtual point-of-sale projects in 2 years. Used Java, Apache Tomcat, Spring, Mercurial, Jira.

Selected Publications

- Flatt M., Derici C. Dybvig R. K., Keep A. et. al. "Rebuilding racket on chez scheme (experience report)", ICFP'19
- Derici C. et. al. "A closed-domain question answering framework using reliable resources to assist students" Natural Language Engineering'18
- Derici C. et. al. "Question analysis for a closed domain question answering system", CICLING'15
- Derici C. et. al. "Rule-based focus extraction in Turkish question answering systems", SIU'14
- Başar R. E., Derici C., and Şenol Ç. "World With Web: A compiler from world applications to JavaScript". Technical Report, Scheme and Functional Programming Workshop'09

Awards & Scholarships

- Scholarship and award for a project on teaching natural languages to hearing impaired, 2014.
- Full Scholarship for PhD, 2015-2025
- Full Scholarship for MSc, 2012
- Full Scholarship for BSc, 2005-2010

Last compiled on February 19, 2026